

i.e., The transresistance amplifier must be designed with a very low input resistance. Ideally $R_i = 0$.

Using voltage division rule at the output we have

$$V_o = \frac{R_M I_i R_L}{R_o + R_L}$$

or

$$V_o = \frac{R_M I_i}{1 + \frac{R_o}{R_L}} \quad (6.21)$$

It is desirable that, the entire output voltage $R_M I_i$ be available across the load

$$\text{i.e.,} \quad V_o \approx R_M I_i \quad (6.22)$$

From Equation (6.21) we find that, the above requirement is satisfied when

$$\frac{R_o}{R_L} \ll 1$$

$$\Rightarrow R_o \ll R_L \quad (6.23)$$

i.e., The transresistance amplifier must be designed with very low output resistance. Ideally $R_o = 0$.

Now we conclude the following:

A good transresistance amplifier must have very low input and output resistances. Ideally $R_i = 0$ and $R_o = 0$.

From Equation (6.22) we have

$$R_M = \frac{V_o}{I_i} \quad (6.24)$$

R_M is called the open loop transresistance or the transresistance without feedback.

The results are summarised in Table 6.1 for ideal amplifiers.

Table 6.1 Summary of ideal amplifier characteristics

Parameter	Voltage amplifier	Current amplifier	Transconductance amplifier	Transresistance amplifier
Input resistance R_i	∞	0	∞	0
Output resistance R_o	0	∞	∞	0

◆ 6.2 FEEDBACK CONCEPT

Several characteristics of the amplifier such as input resistance, output resistance, linearity and band width can be improved by incorporating negative feedback. This can be achieved by feeding

back a part of the output into the input. Such amplifiers are called *feedback amplifiers*. The block diagram of a typical feedback amplifier is shown in Fig. 6.5.

The basic configuration consists of five blocks as shown in Fig. 6.5.

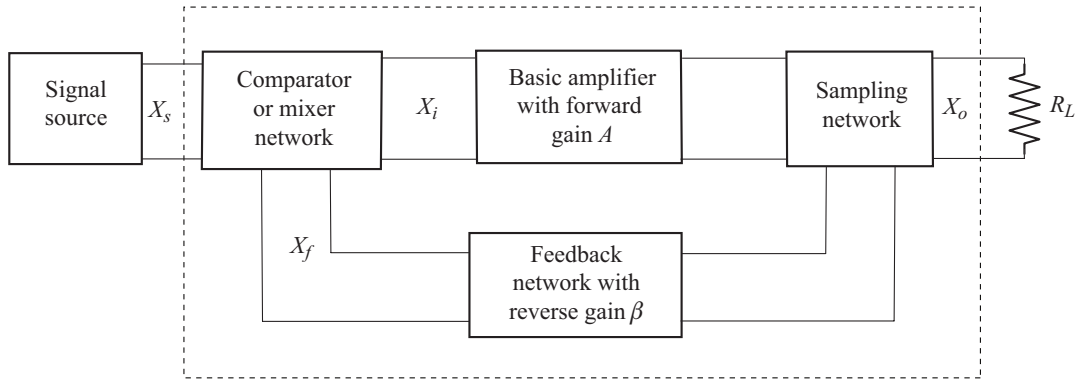


Fig. 6.5 Block diagram of a typical feedback amplifier

X_s : Signal source

X_f : Feedback signal

X_i : Input signal to the basic amplifier

X_o : Output signal

Signal Source Block

The signal source is either a voltage source or a current source depending on the type of amplifier as classified in Section 6.1. A voltage source is represented by a signal source V_s in series with a source resistance R_s , commonly known as Thevenin's representation. A current source is represented by a signal source I_s in parallel with a source resistance R_s , commonly known as Norton's representation. These representations are shown in Fig. 6.6.

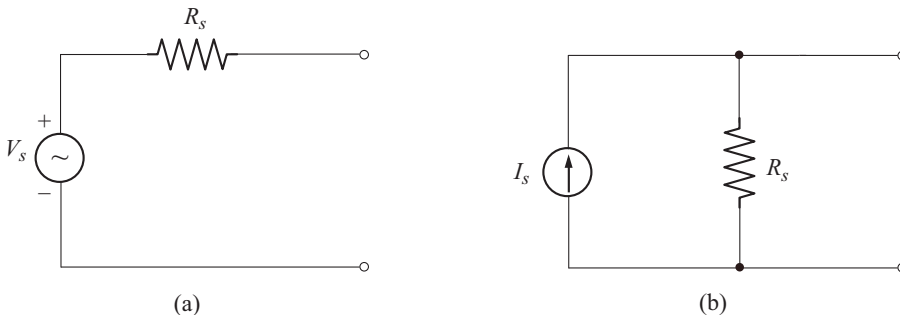


Fig. 6.6 Signal source representation

(a) Thevenin's representation of voltage source

(b) Norton's representation of current source

Comparator or Mixing Block

This block essentially combines the source signal with the feedback signal. The output of the mixer is $X_i = X_s - X_f$. Depending upon the nature of the signal source and the feedback signal there could be either series mixing or shunt mixing. When the source and the feedback signals are both voltages, series mixing is used. If these are both currents, shunt mixing is used. The type of mixing is independent of the output signal being sampled to be feedback.

For instance, an output voltage could be sampled and feedback as an input current or an output current could be sampled and feedback as an input voltage. For example, in the transconductance amplifier, the input is a voltage, whereas the output is a current. Obviously, since the input is a voltage source, only series mixing is possible. Thus, the sampled output current must be converted to a voltage to facilitate series mixing. The schemes for series mixing and shunt mixing are shown in Fig. 6.7.

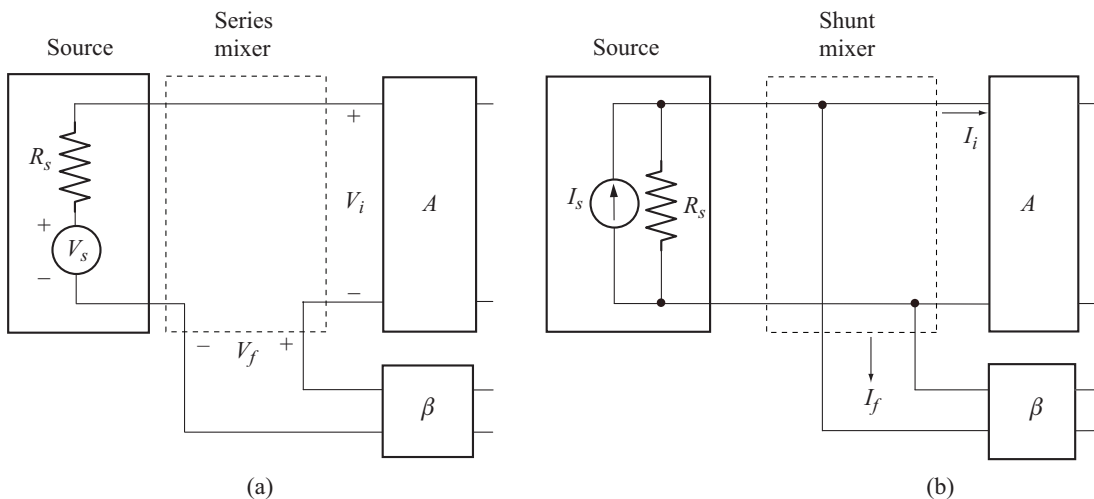


Fig. 6.7 Mixer Block
(a) Series Mixing
(b) Shunt Mixing

Basic Amplifier Block

The basic amplifier block is shown in Fig. 6.8.

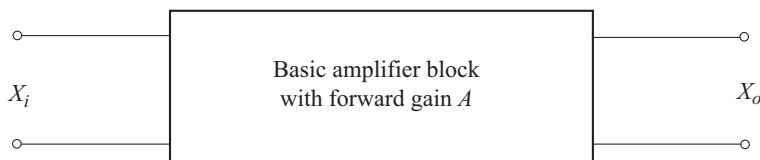


Fig. 6.8 Basic amplifier block

The ratio of the output signal to the input signal of the basic amplifier is represented by A and it is called the transfer gain without feedback or open loop transfer gain or simply open loop gain. It is given by

$$A = \frac{X_o}{X_i} \quad (6.25)$$

A depends on the type of amplifier the block represents.

If it represents a voltage amplifier, then

$$\begin{aligned} X_i &= V_i \quad \text{and} \quad X_o = V_o \\ \therefore A &= A_v = \frac{V_o}{V_i} \end{aligned} \quad (6.26)$$

If the block represents a current amplifier, then

$$\begin{aligned} X_i &= I_i \quad \text{and} \quad X_o = I_o \\ A &= A_i = \frac{I_o}{I_i} \end{aligned} \quad (6.27)$$

Finally if the block represents a transconductance amplifier, then

$$\begin{aligned} X_i &= V_i \quad \text{and} \quad X_o = I_o \\ A &= G_M = \frac{I_o}{V_i} \end{aligned} \quad (6.28)$$

Finally if the block represents a transresistance amplifier, then

$$\begin{aligned} X_i &= I_i \quad \text{and} \quad X_o = V_o \\ A &= R_M = \frac{V_o}{I_i} \end{aligned} \quad (6.29)$$

Though G_M and R_M are not strictly amplifications since the units of output and input are not the same, the quantities A_v , A_i , G_M and R_M are referred to as the transfer gain of the basic amplifier without taking any feedback into consideration. Thus, A represents one of the quantities A_v , A_i , G_M or R_M depending on the type of transfer.

A_f refers to the transfer gain of the amplifier with feedback. A_f is defined as the ratio of the output signal of the amplifier to its input signal. With reference to Fig. 6.5,

$$A_f = \frac{X_o}{X_s} \quad (6.30)$$

for a basic voltage amplifier,

$$A_{vf} = \frac{V_o}{V_s} \quad (6.31)$$

for a basic current amplifier,

$$A_{if} = \frac{I_o}{I_s} \quad (6.32)$$

for a basic transconductance amplifier,

$$G_{Mf} = \frac{I_o}{V_s} \quad (6.33)$$

while for a basic transresistance amplifier,

$$R_{Mf} = \frac{V_o}{I_s} \quad (6.34)$$

Sampling Network

It is used to sample the output signal of the basic amplifier. The feedback network is connected in parallel with the output terminals when voltage is to be sampled as shown in Fig. 6.9(a). The feedback network is connected in series with the output terminals when current is to be sampled as shown in Fig. 6.9 (b).

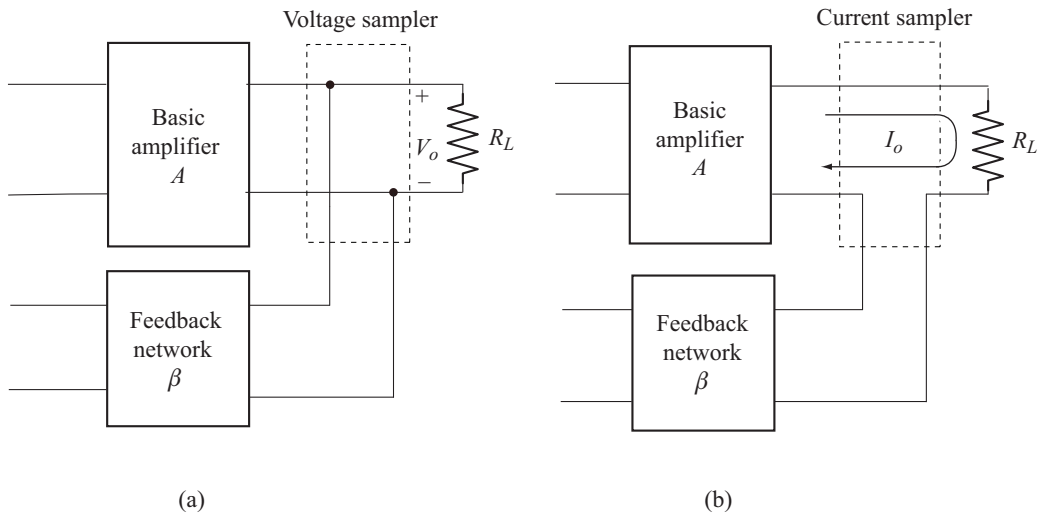


Fig. 6.9 Sampling network
(a) Voltage sampling (b) Current sampling

Feedback Network

This is a passive two-port network configured using passive elements such as resistors, inductors and capacitors. More often, the feedback network is simply a resistive network.

The ratio of the output signal to the input signal of the feedback network is called the feedback factor β .

Feedback factor,
$$\beta = \frac{X_f}{X_o}.$$

Consider the example of sampling a voltage and feeding it back in series.

The feedback signal can be derived from the output voltage by means of a simple voltage divider network as shown in Fig. 6.10.

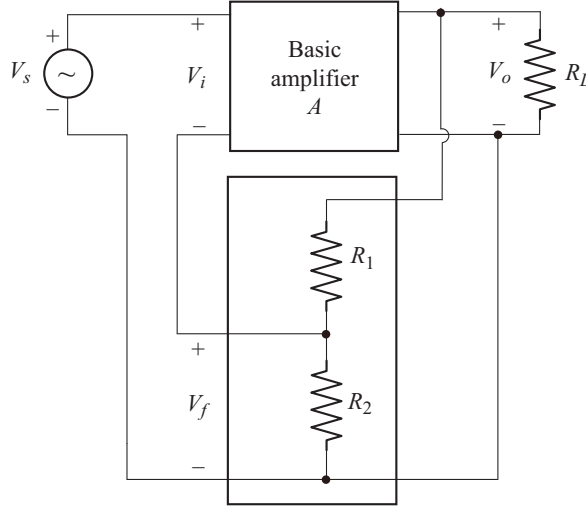


Fig. 6.10 Voltage-sampling and series mixing configuration

The feedback signal or output voltage of the feedback network in Fig. 6.10 is given by

$$V_f = \frac{R_2}{R_1 + R_2} V_o \quad (6.35)$$

$$\therefore \frac{V_f}{V_s} = \frac{R_2}{R_1 + R_2} \quad (6.36)$$

By definition

$$\beta = \frac{V_f}{V_s} = \frac{R_2}{R_1 + R_2} \quad (6.37)$$

Observe that β does not have units in this case. This is not always true, for instance when voltage is sampled and current is derived from the feedback network, the feedback network has the unit of *mhos* or *siemens*.

◆ 6.3 TRANSFER GAIN OF SINGLE LOOP FEEDBACK AMPLIFIER

Let us now, obtain the transfer gain of the feedback amplifier. The block schematic of a single loop feedback amplifier is shown in Fig. 6.11.

In Fig. 6.11 X_s , X_d , X_i , X_o and X_f represent general amplifier signals which may be either voltages or currents depending on the circuit.

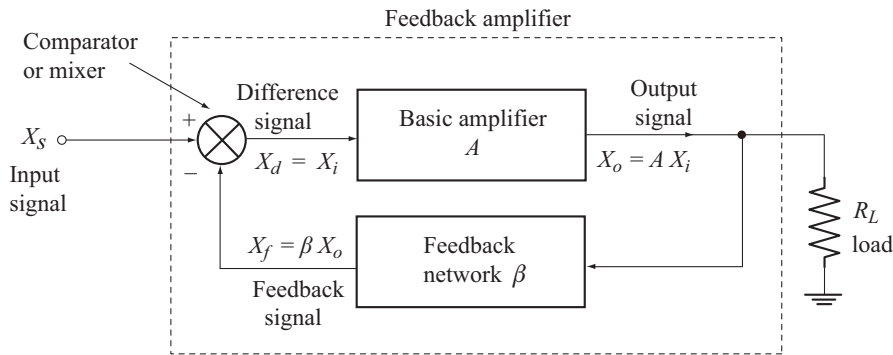


Fig. 6.11 Single-loop feedback amplifier

The transfer gain of the basic amplifier without feedback is given by

$$A = \frac{X_o}{X_i} \quad (6.38)$$

The transfer gain of the feedback amplifier is given by

$$A_f = \frac{X_o}{X_s} \quad (6.39)$$

$$X_o = A X_i \quad (6.40)$$

$$\text{and} \quad X_i = X_s - X_f \quad (6.41)$$

$$\text{where,} \quad X_f = \beta X_o \quad (6.42)$$

Using these relations in Equation (6.40) we have

$$\begin{aligned} X_o &= A (X_s - X_f) = A (X_s - \beta X_o) \\ X_o &= A X_s - A \beta X_o \\ A X_s &= X_o (1 + A \beta) \\ \therefore A_f &= \frac{X_o}{X_s} = \frac{A}{1 + A \beta} \end{aligned} \quad (6.43)$$

Magnitude of the transfer gain is

$$|A_f| = \frac{|A|}{|1 + A \beta|} \quad (6.44)$$

Consider Equation (6.44) :

- If $|1 + A \beta| > 1$, then

$$|A_f| < |A|$$

the feedback is termed negative or degenerative.

- If $|1 + A\beta| < 1$, then

$$|A_f| > |A|$$

the feedback is termed positive or regenerative.

◆ 6.4 LOOP GAIN

Let us now trace the path of X_i through the feedback loop in Fig. 6.11. The signal X_i gets,

- Multiplied by A in the basic amplifier,
- Multiplied by β in the feedback network,
- Multiplied by -1 in the mixer.

Hence, the loop gain or return ratio is $-A\beta$.

Subtracting the loop gain from unity, we get the return difference defined by

$$D = 1 + A\beta \quad (6.45)$$

Using this relation in Equation 6.44 we get

$$|A_f| = \frac{|A|}{|D|} \quad (6.46)$$

Gain with feedback in decibels is

$$20 \log_{10} |A_f| = 20 \log_{10} |A| - 20 \log_{10} |D| \quad (6.47)$$

$$\text{or } 20 \log_{10} |A| = 20 \log_{10} |A_f| + 20 \log_{10} |D| \quad (6.48)$$

This means that in decibels,

$$\text{Gain with out feedback} = \text{gain with feedback} + \text{gain lost due to negative feedback.} \quad (6.49)$$

Thus for instance, a loss of 10 dB gain due to negative feedback is because of the 10 dB feedback negatively in the amplifier.

The amount of feedback in dB = $-\text{gain lost due to negative feedback}$

Let N be the feedback in dB

$$\begin{aligned} \therefore N &= -20 \log_{10} |D| \\ &= 20 \log_{10} \frac{1}{|D|} \end{aligned} \quad (6.50)$$

Substituting for D from Equation 6.46

$$N = -20 \log_{10} \left| \frac{A_f}{A} \right| \quad (6.51)$$

$$\text{or } N = -20 \log_{10} \frac{1}{|1 + A\beta|} \quad (6.52)$$

Example 6.1

A feedback amplifier has a gain of 1000 without feedback. Find the gain with feedback and the amount of feedback in dB for a negative feedback of 10%.

Solution

$$A = 1000$$

$$\% \beta = 10\% \Rightarrow \beta = 0.1$$

From Equation (6.45)

$$D = 1 + A\beta = 1 + 1000 \times 0.1 = 101$$

From Equation (6.46)

$$|A_f| = \frac{|A|}{|D|} = \frac{1000}{101} = 9.9$$

From Equation (6.52)

$$N = 20 \log_{10} \left[\frac{1}{101} \right]$$

$$N = -40 \text{ dB} \quad (\text{A})$$

Gain in dB without feedback is

$$20 \log_{10} |A| = 20 \log_{10} [1000] = 60 \text{ dB} \quad (\text{B})$$

Gain in dB with feedback is

$$20 \log_{10} |A_f| = 20 \log_{10} [9.9] = 20 \text{ dB} \quad (\text{C})$$

Observe that the negative feedback of 40 dB [Equation (A)] has caused the gain to drop from 60 dB to 20 dB [Equation (C)].

◆ 6.5 ASSUMPTIONS IN THE ANALYSIS OF FEEDBACK AMPLIFIERS

Three conditions are to be assumed to simplify the analysis of feedback amplifiers.

Condition 1: The input signal is transmitted to the output only through the forward basic amplifier and not through the feedback network. Thus, if the amplifier is deactivated, the output signal should be zero.

Condition 2: The feedback signal is transmitted from the output to the input only through the feedback network and not through the amplifier.

Condition 3: The feedback factor β is independent of load and source resistances. This can be ensured by proper choice of the elements of the feedback network.

◆ 6.6 FEEDBACK AMPLIFIER TOPOLOGIES

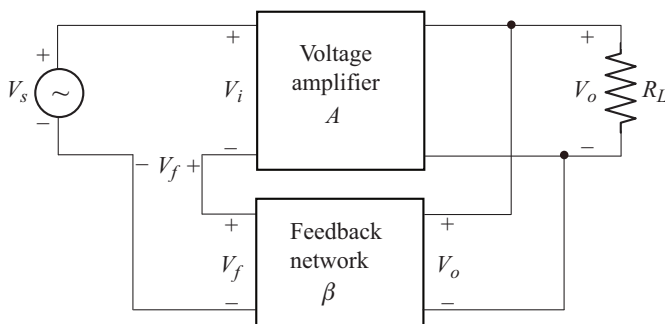
We saw in section 6.2 that there could be two types of sampling, either voltage sampling or current sampling independent of the type of mixing. With each type of sampling there could be either series mixing or shunt mixing. This gives rise to four feedback topologies as shown in Table 6.2.

Table 6.2 Feedback amplifier topologies

<i>Sampling</i>	<i>Mixing</i>	<i>Topology</i>
Voltage	Shunt	Voltage-shunt feedback
Voltage	Series	Voltage-series feedback
Current	Shunt	Current-shunt feedback
Current	Series	Current-series feedback

6.6.1 Voltage-series Feedback Amplifier

Here the output voltage is sampled and the mixing is of series type. For series mixing both source signal and the feedback signal must be voltages. Hence the basic amplifier is a voltage amplifier. Figure 6.12 shows the block diagram of voltage-series feedback amplifier.

**Fig. 6.12** Voltage-series feedback topology

Transfer gain without feedback

$$A = \frac{V_o}{V_i} \quad (6.53)$$

Transfer gain with feedback

$$A_f = \frac{V_o}{V_s} \quad (6.54)$$

It is important to note that

$$A = A_v \quad (6.55)$$

and

$$A_f = A_{vf} \quad (6.56)$$

A_{vf} is the closed loop voltage gain or voltage gain with negative feedback.

Feedback network converts output voltage into feedback voltage. Hence feedback factor is given by

$$\beta = \frac{V_f}{V_o} \quad (6.57)$$

$\Rightarrow$

$$V_f = \beta V_o \quad (6.58)$$

6.6.2 Voltage-shunt Feedback Amplifier

Here the output voltage is sampled and the mixing is of shunt type. For shunt mixing both source signal and the feedback signal must be currents. Hence the basic amplifier is a transresistance amplifier. Figure 6.13 shows the block diagram of voltage-shunt feedback amplifier.

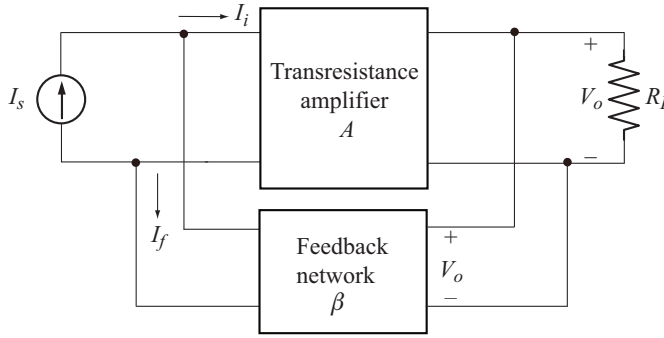


Fig. 6.13 Voltage-shunt feedback topology

Transfer gain without feedback

$$A = \frac{V_o}{I_i} \quad (6.59)$$

Transfer gain with feedback

$$A_f = \frac{V_o}{I_s} \quad (6.60)$$

It is important to note that

$$A = R_M \quad (6.61)$$

and

$$A_f = R_{Mf} \quad (6.62)$$

R_{Mf} is the closed loop transresistance or transresistance with negative feedback.

Feedback network converts output voltage into feedback current. Hence feedback factor is given by

$$\beta = \frac{I_f}{V_o} \quad (6.63)$$

$$\Rightarrow I_f = \beta V_o \quad (6.64)$$

6.6.3 Current-shunt Feedback Amplifier

Here the output current is sampled and the mixing is of shunt type. For shunt mixing both source signal and the feedback signal must be currents. Hence the basic amplifier is a current amplifier. Figure 6.14 shows the block diagram of current-shunt feedback amplifier.

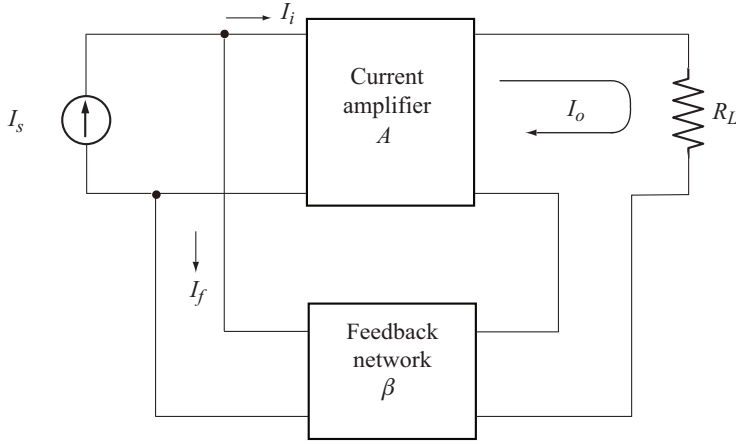


Fig. 6.14 Current-shunt feedback topology

Transfer gain without feedback $A = \frac{I_o}{I_i}$ (6.65)

Transfer gain with feedback $A_f = \frac{I_o}{I_s}$ (6.66)

It is important to note that

$$A = A_I \quad (6.67)$$

and $A_f = A_{If}$ (6.68)

A_{If} is the closed loop current gain or current gain with negative feedback.

Feedback network converts output current into feedback current. Hence the feedback factor is given by

$$\beta = \frac{I_f}{I_o} \quad (6.69)$$

$$\Rightarrow I_f = \beta I_o \quad (6.70)$$

6.6.4 Current-series Feedback Amplifier

Here the output current is sampled and the mixing is of series type. For series mixing both source signal and the feedback signal must be voltages. Hence the basic amplifier is a transconductance amplifier. Figure 6.15 shows the block diagram of current-series feedback amplifier.

Transfer gain without feedback $A = \frac{I_o}{V_i}$ (6.71)

Transfer gain with feedback $A_f = \frac{I_o}{V_s}$ (6.72)

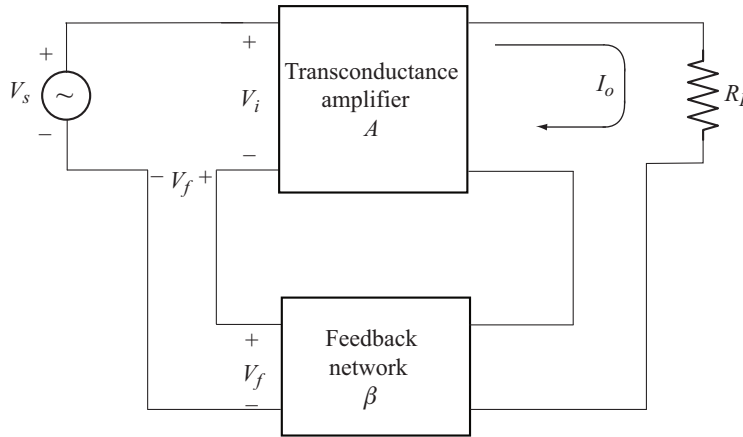


Fig. 6.15 Current-series feedback topology

It is important to note that

$$A = G_M \quad (6.73)$$

and

$$A_f = G_{Mf} \quad (6.74)$$

G_{Mf} is the closed loop transconductance or transconductance with negative feedback.

Feedback network converts output current in to feedback voltage. Hence the feedback factor is given by

$$\beta = \frac{V_f}{I_o} \quad (6.75)$$

$$\Rightarrow V_f = \beta I_o \quad (6.76)$$

Note: While writing the block diagrams of feedback topologies, sources are considered to be ideal. Hence R_s is not shown.

6.6.5 Summary of Feedback Amplifier Topologies

Table 6.3 summarises the feedback amplifier topologies.

Table 6.3 Summary of feedback amplifier topologies

Parameter	Feedback topologies			
	Voltage series	Voltage shunt	Current series	Current shunt
Reference	Fig. 6.12	Fig. 6.13	Fig. 6.15	Fig. 6.14
X_o	Voltage	Voltage	Current	Current
X_s, X_f	Voltage	Current	Voltage	Current
A	$A_v = \frac{V_o}{V_i}$	$R_M = \frac{V_o}{I_i}$	$G_M = \frac{I_o}{V_i}$	$A_I = \frac{I_o}{I_i}$

Parameter	Feedback topologies			
	Voltage series	Voltage shunt	Current series	Current shunt
A_f	$A_{vf} = \frac{V_o}{V_s}$	$R_{Mf} = \frac{V_o}{I_s}$	$G_{Mf} = \frac{I_o}{V_s}$	$A_{if} = \frac{I_o}{I_s}$
β	$\frac{V_f}{V_o}$	$\frac{I_f}{V_o}$	$\frac{V_f}{I_o}$	$\frac{I_f}{I_o}$

Example 6.2

Using the block diagram of voltage series feedback amplifier, derive the expression for its transfer gain with feedback.

Solution

Refer Fig. 6.12

From Equation (6.53), $V_o = A V_i$ (A)

Applying KVL to the input circuit we have

$$V_s - V_i - V_f = 0$$

or $V_i = V_s - V_f$ (B)

Using Equation (B) in Equation (A) we have

$$V_o = A [V_s - V_f]$$

$$= AV_s - AV_f$$
 (C)

But $V_f = \beta V_o$ [From Equation 6.58]

Now $V_o = AV_s - A\beta V_o$

$$V_o [1 + A\beta] = A V_s$$

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A}$$
 (D)

For voltage series feedback amplifier

$$A = A_v \quad \text{and} \quad A_f = A_{vf}$$

Now Equation (D) can also be written as

$$A_{vf} = \frac{A_v}{1 + \beta A_v}$$
 (E)

Example 6.3

Using the block diagram of voltage shunt feedback amplifier, derive the expression for its transfer gain with feedback.

Solution

Refer Fig. 6.13

From Equation (6.59), $V_o = A I_i$ (A)

Applying KCL at the input circuit we have

$$\begin{aligned} I_s &= I_i + I_f \\ \text{or} \quad I_i &= I_s - I_f \end{aligned} \quad (B)$$

Using Equation (B) in Equation (A) we have

$$\begin{aligned} V_o &= A [I_s - I_f] \\ &= A I_s - A I_f \end{aligned} \quad (C)$$

But $I_f = \beta V_o$ [From Equation 6.64]

Now $V_o = A I_s - A \beta V_o$

$$V_o [1 + A \beta] = A I_s$$

$$A_f = \frac{V_o}{I_s} = \frac{A}{1 + A \beta} \quad (D)$$

For voltage shunt feedback amplifier

$$A = R_M \quad \text{and} \quad A_f = R_{Mf}$$

Now Equation (D) can also be written as

$$R_{Mf} = \frac{R_M}{1 + \beta R_M} \quad (E)$$

Note: Similarly we can derive the expressions for A_f of current series and current shunt feedback amplifiers using the block diagrams given in Fig. 6.15 and Fig. 6.14 respectively.

◆ 6.7 GENERAL CHARACTERISTICS OF NEGATIVE FEEDBACK AMPLIFIER

Though negative feedback reduces the transfer gain it is extensively used in amplifiers. The reason is that with negative feedback, many desirable characteristics are obtained. Now let us examine these desirable characteristics in detail.

6.7.1 Stability of Transfer Gain

Transfer gain with feedback is given by

$$A_f = \frac{A}{1 + \beta A} \quad (6.77)$$

Differentiating with respect to A , we have

$$\frac{dA_f}{dA} = \frac{[1 + \beta A] - A [\beta]}{[1 + \beta A]^2}$$

$$\begin{aligned}
&= \frac{1}{[1 + \beta A]^2} \\
&= \frac{1}{[1 + \beta A]} \cdot \frac{1}{[1 + \beta A]}
\end{aligned} \tag{6.78}$$

From Equation (6.77) we have

$$\frac{1}{1 + \beta A} = \frac{A_f}{A}$$

Using this relation in Equation (6.78) we have

$$\begin{aligned}
\frac{dA_f}{dA} &= \frac{A_f}{A} \cdot \frac{1}{1 + \beta A} \\
\frac{dA_f}{dA} &= \frac{\left[\frac{dA}{A} \right]}{[1 + \beta A]} \\
\left| \frac{dA_f}{A_f} \right| &= \frac{\left| \frac{dA}{A} \right|}{|1 + \beta A|}
\end{aligned} \tag{6.79}$$

$$\left| \frac{dA_f}{A_f} \right| = \text{relative change in gain with feedback}$$

$$\left| \frac{dA}{A} \right| = \text{relative change in gain without feedback}$$

For negative feedback, $|1 + \beta A| > 1$

Now from Equation (6.79) we get

$$\left| \frac{dA_f}{A_f} \right| < \left| \frac{dA}{A} \right| \tag{6.80}$$

or stated in words:

Relative change in gain with feedback is less than that without feedback. This implies that with negative feedback, the transfer gain changes very little or negative feedback stabilises the transfer gain.

Desensitivity factor

From Equation (6.79) we find that the variation in transfer gain or the sensitivity of transfer gain gets reduced by a factor $D = 1 + \beta A$. For this reason, D is also called the Desensitivity factor.

Example 6.4

An amplifier without feedback has a transfer gain of -2000 . The transfer gain changes by 20% due to temperature. If negative feedback with $\beta = -0.1$ is given, calculate the change in gain of the feedback amplifier.

Solution

$$\begin{aligned}
 A &= -2000 \quad \beta = -0.1 \\
 \% \left| \frac{dA}{A} \right| &= 20\% \\
 1 + \beta A &= 1 + (-0.1)(-2000) \\
 &= 201
 \end{aligned}$$

From Equation (6.79)

$$\begin{aligned}
 \% \left| \frac{dA_f}{A_f} \right| &= \frac{\% \left| \frac{dA}{A} \right|}{1 + \beta A} \\
 &= \frac{20\%}{201} = 0.099\%
 \end{aligned}$$

Example 6.5

Calculate the gain of a negative-feedback amplifier having $A = -2000$, if the feedback factor is 20% .

Solution

$$\begin{aligned}
 A &= -2000 \\
 \% \beta &= 20\% \\
 \Rightarrow \beta &= 0.2
 \end{aligned}$$

For negative feedback, $A\beta$ is positive. Therefore we take

$$\begin{aligned}
 \beta &= -0.2 \\
 1 + \beta A &= 1 + (-0.2)(-2000) \\
 &= 401 \\
 A_f &= \frac{A}{1 + \beta A} \\
 &= \frac{-2000}{401} = -4.98
 \end{aligned}$$

6.7.2 Reduction in Non-linear Distortion

The transfer characteristics of a practical amplifier is non-linear as shown in Fig. 6.16.

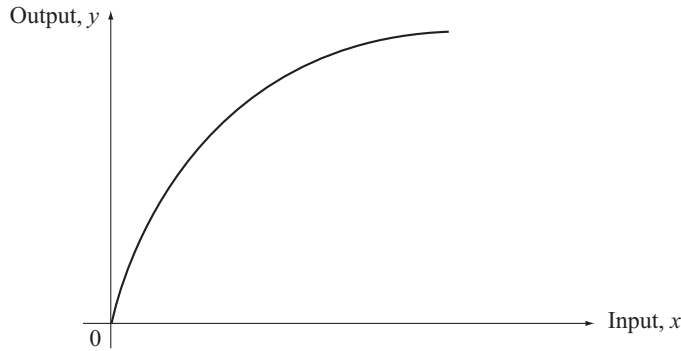


Fig. 6.16 Transfer characteristics of practical amplifier

Since, the transfer characteristics is non-linear, the input-output relationship can be expressed as

$$y = k_1 x + k_2 x^2 + k_3 x^3 + \dots \quad (6.81)$$

where the first term represents the linear term followed by the non-linear terms. For instance, if the input is sinusoidal of large amplitude, say

$$x = A_o \sin \omega t \quad (6.82)$$

Then, Equation (6.81) becomes

$$y = k_1 A_o \sin \omega t + k_2 A_o^2 \sin^2 \omega t + k_3 A_o^3 \sin^3 \omega t + \dots$$

which can be expressed in the general form as

$$y = B_o + B_1 \sin \omega t + B_2 \sin 2 \omega t + B_3 \sin 3 \omega t + \dots \quad (6.83)$$

where B_o is the dc term, B_1 is the amplitude of the fundamental, B_2 is the amplitude of the second harmonic and so on.

Equation 6.83 indicates that y is not an exact replace of x , but is distorted, referred to as harmonic distortion or non-linear distortion. It can be shown from Fourier analysis that as the order of the harmonic increases, its amplitude decreases.

$$\text{i.e.,} \quad |B_1| > |B_2| > |B_3| > \dots \quad (6.84)$$

To study the effect of negative feedback on harmonic distortion, let us assume that only second harmonic distortion is present. Let B_{2f} be the second harmonic distortion at the output in the presence of negative feedback.

When B_{2f} is sampled it gets

- Multiplied by β in the feedback network
- Multiplied by -1 in the mixer
- Multiplied by A in the basic amplifier

Finally it becomes $-A\beta B_{2f}$

Hence, the second harmonic component at the output in the presence of negative feedback is

$$B_{2f} = B_2 - A\beta B_{2f}$$

$$B_{2f}(1 + A\beta) = B_2$$

$$\therefore B_{2f} = \frac{B_2}{1 + A\beta} = \frac{B_2}{D} \quad (6.85)$$

$$\text{or} \quad |B_{2f}| = \frac{|B_2|}{|1 + A\beta|} = \frac{|B_2|}{|D|} \quad (6.86)$$

We know that for negative feedback

$$|1 + A\beta| > 1 \quad \text{and hence} \quad |B_{2f}| < |B_2|$$

Observe that the second-harmonic distortion gets reduced by a factor $|1 + A\beta|$.

6.7.3 Reduction of Noise

Let N represents the output noise without feedback and let N_f represents the output noise with feedback. When N_f is sampled, it gets multiplied by $-A\beta$ around the feedback loop and becomes $-A\beta N_f$.

Thus, the noise at the output in the presence of negative feedback is

$$\begin{aligned} N_f &= N - A\beta N_f \\ \text{or} \quad N_f &= \frac{N}{1 + A\beta} = \frac{N}{D} \end{aligned} \quad (6.87)$$

$$\text{or} \quad |N_f| = \frac{|N|}{|1 + A\beta|} = \frac{|N|}{|D|} \quad (6.88)$$

Equation (6.88) shows that the output noise reduces by a factor of $|1 + A\beta|$ in the presence of negative feedback since $|1 + A\beta| > 1$.

6.7.4 Reduction in Frequency Distortion

The gain of an amplifier becomes a function of frequency due to internal device capacitances, associated coupling and bypass elements such as capacitors and transformers. As a result, different frequency components are amplified to different extents.

We have seen that the gain of an amplifier with negative feedback is given by

$$A_f = \frac{A}{1 + A\beta}$$

$$\text{If } |A\beta| \gg 1, \text{ then} \quad A_f \approx \frac{A}{A\beta}$$

$$\therefore A_f = \frac{1}{\beta} \quad (6.89)$$

Under this condition, A_f depends only on β . In most of the circuits, β network contains only resistors and therefore the feedback factor β is independent of frequency. As a result, A_f becomes independent of frequency. Thus negative feedback reduces frequency distortion.

◆ 6.8 EFFECT OF NEGATIVE FEEDBACK ON INPUT RESISTANCE

The effect of negative feedback on input resistance depends on the nature of mixing.

In case of series mixing the input resistance increases while in the case of shunt mixing the input resistance decreases.

Let us now see how series mixing increases the input resistance. The input stage of a feedback amplifier with series mixing is shown in Fig. 6.17.

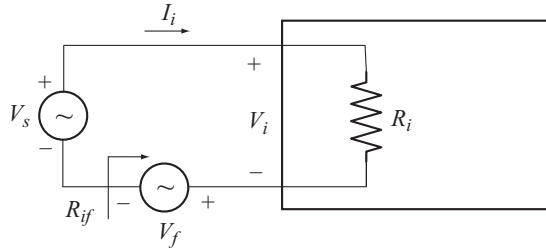


Fig. 6.17 Input stage of feedback amplifier with series mixing

From Fig. 6.17 the input resistance without feedback is given by

$$R_i = \frac{V_i}{I_i} \quad (6.90)$$

While the input resistance with feedback R_{if} is given by

$$R_{if} = \frac{V_s}{I_i} \quad (6.91)$$

In the presence of negative feedback V_f is in opposite polarity with respect to V_s and hence I_i with negative feedback is less than that without feedback. As a result R_{if} is greater than R_i .

Now, let us look at the situation in case of shunt mixing. The input stage of a feedback amplifier with shunt mixing is shown in Fig. 6.18.

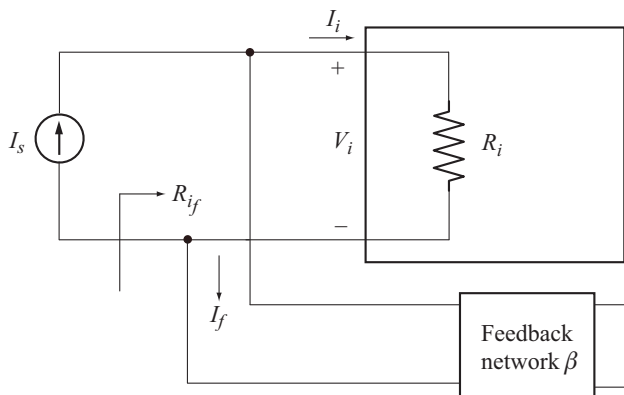


Fig. 6.18 Input stage of feedback amplifier with shunt mixing

Observe that the feedback network appears in parallel with the input terminals of the amplifier. From Fig. 6.18 input resistance without feedback is

$$R_i = \frac{V_i}{I_i} \quad (6.92)$$

and in the presence of negative feedback, the input resistance is

$$R_{if} = \frac{V_i}{I_s} \quad (6.93)$$

But

$$I_s = I_i + I_f$$

$$R_{if} = \frac{V_i}{I_i + I_f} \quad (6.94)$$

From Equations (6.92) and (6.94) it is clear that $R_{if} < R_i$.

6.8.1 Input Resistance of Voltage-series Feedback Amplifier

We now obtain an expression for R_{if} for the voltage-series feedback amplifier. Figure 6.19 shows a typical voltage-series feedback circuit.

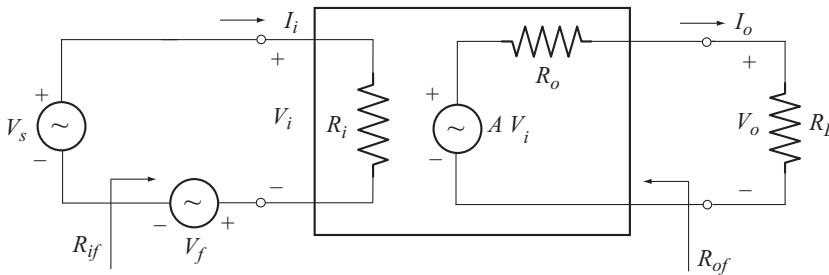


Fig. 6.19 Voltage series feedback amplifier

$$R_i = \frac{V_i}{I_i} = \text{Input resistance without feedback}$$

$$R_{if} = \frac{V_s}{I_i} = \text{Input resistance with feedback.}$$

Applying KVL to the input circuit of Fig. 6.19, we have

$$V_s - V_i - V_f = 0 \quad (6.95)$$

$$\text{But } V_i = I_i R_i \text{ and } V_f = \beta V_o$$

Substituting these relations in Equation 6.95 we have

$$V_s - I_i R_i - \beta V_o = 0 \quad (6.96)$$

$$\text{But } V_o = A V_i = A I_i R_i$$

Using this relation in Equation (6.96) we have

$$\begin{aligned}
 V_s - I_i R_i - \beta A I_i R_i &= 0 \\
 V_s &= I_i R_i [1 + \beta A] \\
 \text{Now } R_{if} &= \frac{V_s}{I_i} = R_i [1 + \beta A] \\
 \text{Since } [1 + \beta A] &> 1 \\
 R_{if} &> R_i
 \end{aligned} \tag{6.97}$$

Note that the input resistance increases by a factor $1 + \beta A$ as expected for series mixing.

6.8.2 Input Resistance of Voltage-shunt Feedback Amplifier

The input resistance can be computed from the voltage-series feedback circuit shown in Fig. 6.20.

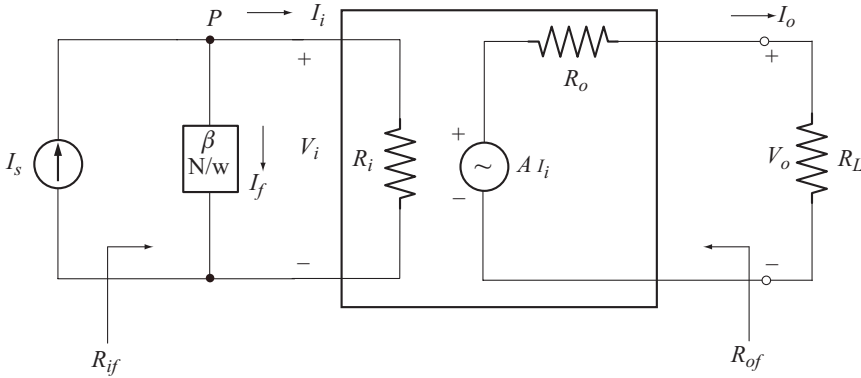


Fig. 6.20 Voltage - shunt feedback amplifier

$$R_i = \frac{V_i}{I_i} = \text{Input resistance without feedback}$$

$$R_{if} = \frac{V_i}{I_s} = \text{Input resistance with feedback}$$

Applying KCL at the node P of Fig. 6.20

$$I_s = I_i + I_f \tag{6.98}$$

$$\text{But } I_i = \frac{V_i}{R_i} \text{ and } I_f = \beta V_o$$

Substituting these relations in Equation (6.98) we have

$$I_s = \frac{V_i}{R_i} + \beta V_o$$

$$\begin{aligned}\text{Also } V_o &= A I_i \\ &= A \frac{V_i}{R_i}\end{aligned}$$

Using this relation in the above Equation we get

$$\begin{aligned}I_s &= \frac{V_i}{R_i} + \beta A \frac{V_i}{R_i} \\ &= \frac{V_i}{R_i} [1 + \beta A]\end{aligned}$$

$$\text{Now } R_{if} = \frac{V_i}{I_s} = \frac{R_i}{1 + \beta A} \quad (6.99)$$

$$\text{Since } [1 + \beta A] > 1$$

$$R_{if} < R_i$$

Note that the input resistance decreases by a factor $1 + \beta A$ as expected for shunt mixing.

6.8.3 Input Resistance of Current-series Feedback Amplifier

The circuit for computation of the input resistance of a current-series feedback amplifier is shown in Fig. 6.21.

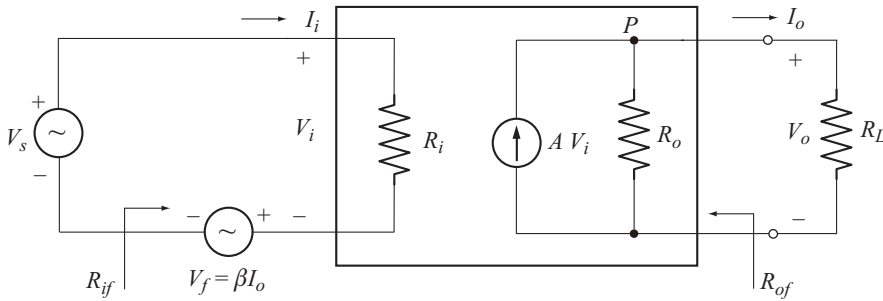


Fig. 6.21 Current-series feedback amplifier

$$R_i = \frac{V_i}{I_i} = \text{Input resistance without feedback}$$

$$R_{if} = \frac{V_s}{I_i} = \text{Input resistance with feedback}$$

Applying KVL to the input circuit of Fig. 6.21, we have

$$V_s - V_i - V_f = 0 \quad (6.100)$$

$$\text{But } V_i = I_i R_i \text{ and } V_f = \beta I_o$$

Using these relations in Equation (6.100) we have

$$V_s - I_i R_i - \beta I_o = 0 \quad (6.101)$$

$$\begin{aligned} \text{But } I_o &= A V_i \\ &= A I_i R_i \end{aligned}$$

Using this relation in Equation (6.101) we get

$$\begin{aligned} V_s - I_i R_i - \beta A I_i R_i &= 0 \\ V_s &= I_i R_i [1 + \beta A] \\ \text{Now } R_{if} &= \frac{V_s}{I_i} = R_i [1 + \beta A] \end{aligned} \quad (6.102)$$

$$\begin{aligned} \text{Since } [1 + \beta A] &> 1 \\ R_{if} &> R_i \end{aligned}$$

Note that the input resistance increases by a factor $1 + \beta A$ as expected for series mixing.

6.8.4 Input Resistance of Current-shunt Feedback Amplifier

Figure 6.22 shows the circuit of a current-shunt feedback amplifier for the computation of input resistance.

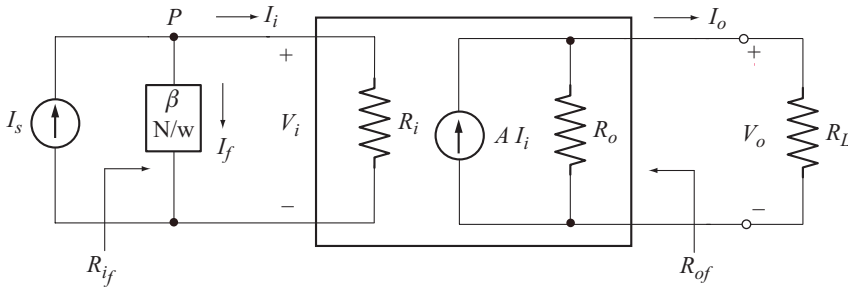


Fig. 6.22 Current-shunt feedback amplifier

$$R_i = \frac{V_i}{I_i} = \text{Input resistance without feedback,}$$

$$R_{if} = \frac{V_i}{I_s} = \text{Input resistance with feedback.}$$

Applying KCL to the input node P we have

$$I_s = I_i + I_f \quad (6.103)$$

$$\text{But } I_i = \frac{V_i}{R_i} \text{ and } I_f = \beta I_o$$

Substituting these relations in Equation (6.103) we have

$$I_s = \frac{V_i}{R_i} + \beta I_o \quad (6.104)$$

But

$$\begin{aligned} I_o &= A I_i \\ &= A \frac{V_i}{R_i} \end{aligned}$$

Using this relations in Equation (6.104) we have

$$\begin{aligned} I_s &= \frac{V_i}{R_i} + \beta A \frac{V_i}{R_i} \\ &= \frac{V_i}{R_i} [1 + \beta A] \end{aligned}$$

Now

$$R_{if} = \frac{V_i}{I_s} = \frac{R_i}{1 + \beta A} \quad (6.105)$$

Since $[1 + \beta A] > 1$

$$R_{if} < R_i$$

Note that input resistance decreases by a factor $1 + \beta A$ for shunt mixing.

◆ 6.9 EFFECT OF NEGATIVE FEEDBACK ON OUTPUT RESISTANCE

The effect of negative feedback on output resistance depends on the nature of sampling.

The output resistance decreases in the case of voltage sampling while it increases in the case of the current sampling.

Let us first see how voltage sampling reduces the output resistance. Consider the output circuit of a typical voltage amplifier shown in Fig. 6.23.

We know that,

$$A_f = \frac{A}{1 + \beta A}$$

If $\beta A \gg 1$

Then

$$A_f \approx \frac{A}{\beta A} = \frac{1}{\beta}$$

Note that A_f is independent of both source and load resistances. Thus, ideally the output voltage V_o would remain the same irrespective of changes in load resistance R_L in the presence of negative feedback. This is possible in the circuit of Fig. 6.23, only if

$$R_{of} \approx 0 \quad \text{or} \quad R_{of} \ll R_L$$

Under this condition,

$$V_o = A_f V_s$$

The circuit to analyse the effect of negative feedback on a current sampling circuit is shown in Fig. 6.24.

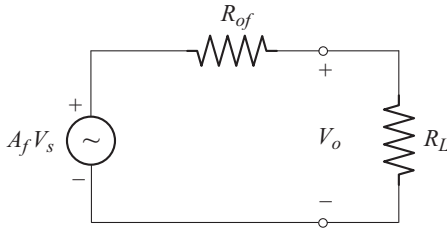


Fig. 6.23 Output stage of a feedback amplifier with voltage sampling

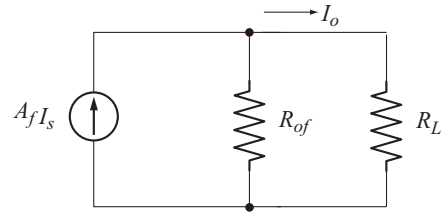


Fig. 6.24 Output stage of a feedback amplifier with current sampling

In this case, under stabilization through negative feedback the current I_o through R_L remain unchanged with changes in R_L , since A_f is independent of source and load resistances under negative feedback. This is obviously possible only if $R_{of} = \infty$ or $R_{of} \gg R_L$ in the circuit of Fig. 6.24, in which case

$$I_o = A_f I_s$$

6.9.1 Output Resistance of Voltage-Series Feedback Amplifier

We apply the following procedure to find the output resistance of feedback amplifier

1. Reduce V_s (I_s) to zero
2. Remove R_L and connect a voltage source V between the output terminals.

Let I be the current driven by the voltage source V into the amplifier output terminals. The output resistance with feedback is given by

$$R_{of} = \frac{V}{I} \quad (6.106)$$

Let us apply these steps to the voltage series feedback amplifier of Fig. 6.19. The resulting circuit is shown in Fig. 6.25.

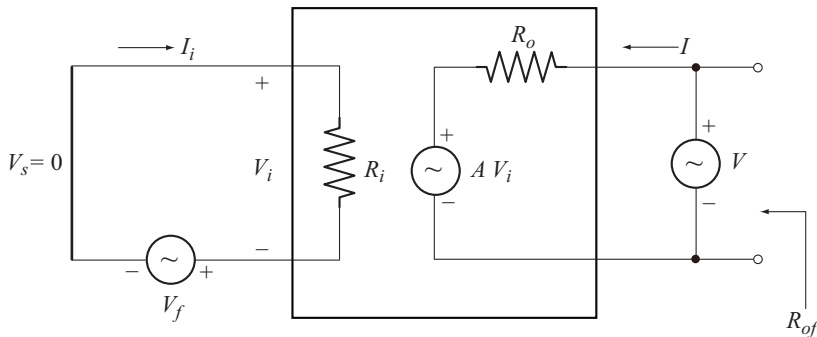


Fig. 6.25 Output resistance of voltage series feedback amplifier

Applying KVL to the input circuit we have

$$V_s - V_i - V_f = 0 \quad (6.107)$$

Since $V_s = 0$, we get

$$V_i = -V_f \quad (6.108)$$

$$V_f = \beta V_o$$

But $V_o = V$

$$\therefore V_f = \beta V$$

Using this relation in Equation (6.108) we get

$$V_i = -\beta V \quad (6.109)$$

Applying KVL to the output circuit, we have

$$V - I R_o - A V_i = 0 \quad (6.110)$$

Substituting Equation (6.109) in Equation (6.110) we have

$$V - I R_o - A [-\beta V] = 0$$

$$V [1 + \beta A] = I R_o$$

$$\text{Now } R_{of} = \frac{V}{I} = \frac{R_o}{1 + \beta A} \quad (6.111)$$

Since, $(1 + \beta A) > 1$

$$R_{of} < R_o$$

Note that the output resistance decreases by a factor $1 + \beta A$ as expected for voltage sampling.

6.9.2 Output Resistance of Voltage Shunt Feedback Amplifier

Let us obtain the equivalent circuit shown in Fig. 6.26 from the circuit of Fig. 6.20 by

1. Reducing I_s to zero,
2. Replacing R_L by a voltage source V .

Applying KCL at the node P of the circuit of Fig. 6.26

$$I_i + I_f = 0$$

$$\therefore I_i = -I_f \quad (6.112)$$

But $I_f = \beta V_o = \beta V$

$$\therefore I_i = -\beta V \quad (6.113)$$

Applying KVL to the output circuit of Fig. 6.26

$$A I_i + I R_o - V = 0 \quad (6.114)$$

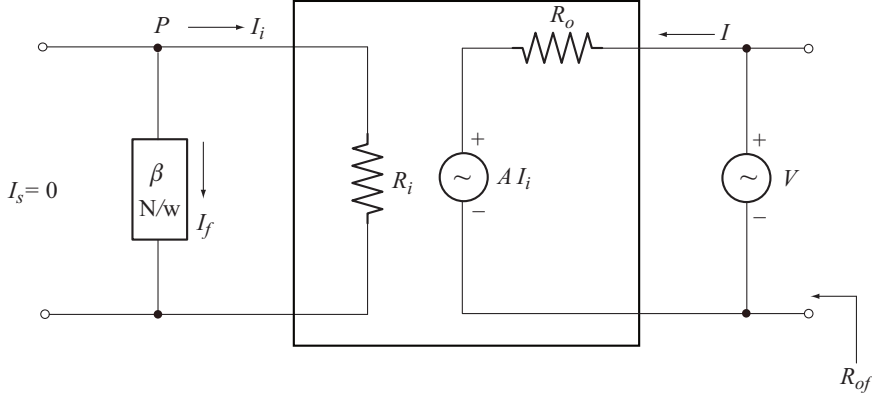


Fig. 6.26 Output resistance of voltage shunt feedback amplifier

Substituting for I_i from Equation (6.113)

$$A(-\beta V) + I R_o - V = 0$$

$$I R_o = V(1 + A\beta) \quad (6.115)$$

By definition

$$R_{of} = \frac{V}{I}$$

$\therefore$ From Equation (6.115)

$$R_{of} = \frac{R_o}{1 + \beta A} \quad (6.116)$$

Since $[1 + \beta A] > 1$

$$R_{of} < R_o$$

Note that the output resistance decreases by a factor $1 + \beta A$ as expected for voltage sampling.

6.9.3 Output Resistance of Current Series Feedback Amplifier

Let us obtain the equivalent circuit of the current series feedback amplifier shown in Fig. 6.21 to calculate the output resistance. The equivalent circuit is shown in Fig. 6.27.

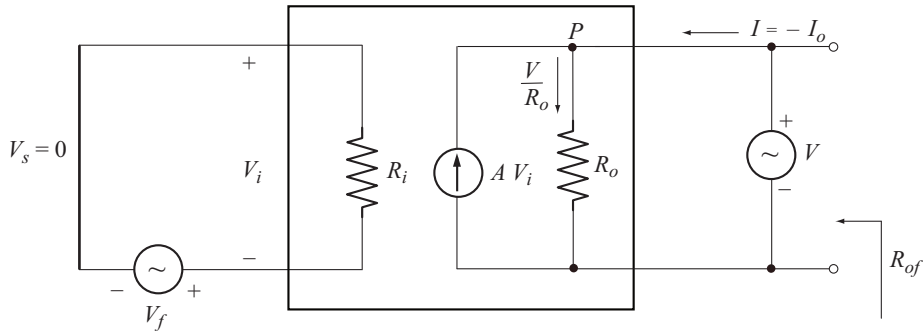


Fig. 6.27 Output resistance of current series feedback amplifier

Applying KVL to the input circuit of Fig. 6.27, we have

$$-V_i - V_f = 0 \quad (6.117)$$

$$\text{or} \quad V_i = -V_f$$

By definition

$$V_f = \beta I_o$$

But

$$I_o = -I \quad (6.118)$$

$$\therefore V_f = -\beta I$$

Substituting this relation in Equation 6.117, we have

$$V_i = \beta I \quad (6.119)$$

Applying KCL at node P of the circuit in Fig. 6.27

$$A V_i + I - \frac{V}{R_o} = 0 \quad (6.120)$$

Substituting for V_i from Equation (6.119)

$$\begin{aligned} A \beta I + I &= \frac{V}{R_o} \\ \frac{V}{R_o} &= I(1 + A \beta) \end{aligned} \quad (6.121)$$

By definition,

$$\begin{aligned} R_{of} &= \frac{V}{I} \\ \therefore R_{of} &= R_o(1 + \beta A) \end{aligned} \quad (6.122)$$

$$\text{Since } [1 + \beta A] > 1$$

$$R_{of} > R_o$$

Note that output resistance increases by a factor $1 + \beta A$ as expected for current sampling.

6.9.4 Output Resistance of Current-shunt Feedback Amplifier

Let us obtain the equivalent circuit of the current-shunt feedback amplifier shown in Fig. 6.22 to calculate the output resistance. The equivalent circuit is shown in Fig. 6.28.

Applying KCL at node P of the input circuit shown in Fig. 6.28, we have

$$I_i + I_f = 0 \quad (6.123)$$

$$\text{or} \quad I_i = -I_f$$

$$\text{By definition} \quad I_f = \beta I_o$$

$$\text{But} \quad I_o = -I$$

$$\therefore I_f = -\beta I \quad (6.124)$$

Substituting this relation in Equation (6.123), we have

$$I_i = \beta I \quad (6.125)$$

Applying KCL at node Q of the output circuit in Fig. 6.28, we have.

$$A I_i + I - \frac{V}{R_o} = 0$$

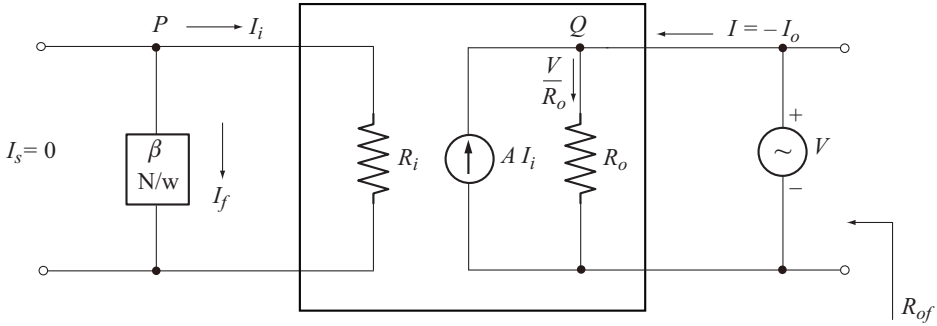


Fig. 6.28 Output resistance of current shunt feedback amplifier

Substituting for I_i from Equation (6.125)

$$A \beta I + I = \frac{V}{R_o} \quad (6.126)$$

$$I(1 + \beta A) = \frac{V}{R_o}$$

$$\text{By definition,} \quad R_{of} = \frac{V}{I}$$

$\therefore$ From Equation (6.126)

$$R_{of} = R_o (1 + \beta A) \quad (6.127)$$

Since $[1 + \beta A] > 1$

$$R_{of} > R_o$$

Note that the output resistance increases by a factor $1 + \beta A$ as expected for current sampling.

Table 6.4 Summarizes the effect of negative feedback on input and output resistances on all the feedback topologies.

Table 6.4 Summary of effect of negative feedback on input and output resistance

<i>Parameter</i>	<i>Feedback topologies</i>			
	<i>Voltage series</i>	<i>Voltage shunt</i>	<i>Current series</i>	<i>Current shunt</i>
R_{if}	$R_i (1 + \beta A)$	$\frac{R_i}{1 + \beta A}$	$R_i (1 + \beta A)$	$\frac{R_i}{1 + \beta A}$
R_{of}	$\frac{R_o}{1 + \beta A}$	$\frac{R_o}{1 + \beta A}$	$R_o (1 + \beta A)$	$R_o (1 + \beta A)$

◆ 6.10 EFFECT OF NEGATIVE FEEDBACK ON BANDWIDTH

For any amplifier the product of gain and bandwidth called the gain-bandwidth product is a constant whether the feedback is present or not.

$\therefore$ Gain-bandwidth product with feedback = Gain-bandwidth product without feedback

$$[A_f][B W_f] = [A][B W]$$

or $B W_f = \frac{A}{A_f} B W$ (6.128)

where $B W$ = Bandwidth without feedback

$B W_f$ = Band width with feedback

Since $A_f = \frac{A}{1 + \beta A}$

We have $\frac{A}{A_f} = 1 + \beta A$

Using this relation in Equation (6.128) we have

$$B W_f = B W [1 + \beta A]$$

Since $[1 + \beta A] > 1$

$$B W_f > B W$$
(6.129)

Note that bandwidth increases by a factor $[1 + \beta A]$ since the gain reduces by the same factor, in order to keep the gain bandwidth product constant.

Figure 6.29 shows the effect of negative feedback on bandwidth and gain.

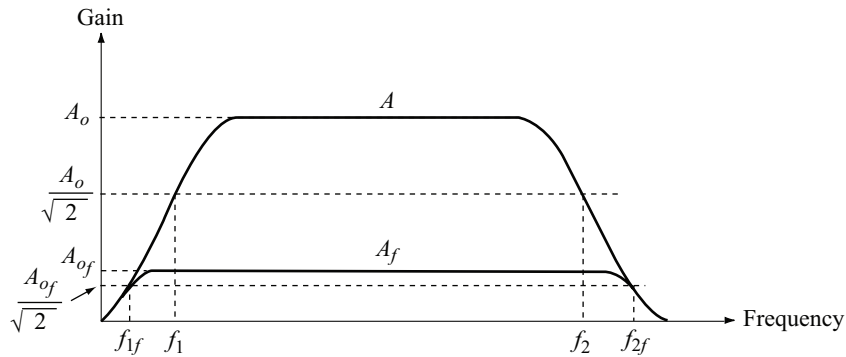


Fig. 6.29 Effect of negative feedback on gain and band width

$$A_o = \text{Midband gain without feedback}$$
$$A_{of} = \text{Midband gain with feedback}$$

$$A_{of} = \frac{A_o}{1 + \beta A_o} \quad (6.130)$$

$$BW_f = f_{2f} - f_{1f} \quad (6.131)$$

$$BW = f_2 - f_1 \quad (6.132)$$

f_{2f} and f_{1f} are the upper and lower 3 dB frequencies with feedback. f_2 and f_1 are the corresponding frequencies without feedback.

Observe that

$$f_{1f} < f_1$$

and $f_{2f} > f_2$

◆ 6.11 ADVANTAGES OF NEGATIVE FEEDBACK

The advantages of negative feedback can now be listed based on the discussions in the preceding sections.

- Negative feedback stabilizes transfer gain.
- Negative feedback reduces non-linear distortion by a factor, $1 + \beta A$. This means that the operation of the amplifier becomes more linear than without feedback.
- Noise output is reduced by a factor, $1 + \beta A$.
- Negative feedback reduces frequency distortion.
- Voltage amplifiers generally have a high input resistance and a low output resistance. Negative feedback further increases the input resistance and further decreases the output resistance for a voltage amplifier.

Similarly, for other topologies, the input and output resistances improve towards their ideal values.

- Negative feedback improves the frequency response of an amplifier.

It is to be noted that all these advantages are derived at the cost of transfer gain as we have seen that the transfer gain with feedback is less than the transfer gain with out feedback.

Example 6.6

An amplifier has a bandwidth of 200 kHz and a voltage gain of 1000.

- What will be the new bandwidth and gain if 5 % negative feedback is introduced.
- What is the gain-bandwidth product with and without feedback.
- What should be the amount of feedback if the bandwidth required is 1 MHz.

Solution

Bandwidth without feedback $BW = 200 \text{ kHz}$.

Voltage gain without feedback $A = 1000$

$$\begin{aligned} \text{(a)} \quad \% \beta &= 5\% \Rightarrow \beta = 0.05 \\ 1 + \beta A &= 1 + (1000)(0.05) = 51 \end{aligned}$$

$\therefore$ Bandwidth with feedback

$$\begin{aligned} BW_f &= BW \times (1 + \beta A) \\ &= (200 \text{ kHz})(51) = 10200 \text{ kHz.} \\ BW_f &= 10.2 \text{ MHz} \end{aligned}$$

(b) Gain bandwidth product without feedback is

$$(A)(BW) = (1000)(200 \text{ kHz}) = 2 \times 10^8$$

$$A_f = \frac{A}{1 + \beta A} = \frac{1000}{51} = 19.6$$

Gain bandwidth product with feedback is

$$(A_f)(BW_f) = (19.6)(10.2 \times 10^6) = 2 \times 10^8$$

The gain bandwidth product is the same with and without feedback as expected

(c) Given $BW_f = 1 \text{ MHz}$

$$\begin{aligned} BW_f &= BW [1 + \beta A] \\ \therefore 1 + \beta A &= \frac{BW_f}{BW} = \frac{1 \text{ MHz}}{200 \text{ kHz}} = 5 \\ \text{Now } \beta &= \frac{5-1}{A} = \frac{4}{1000} = 0.004 \text{ or } 0.4\% \end{aligned}$$

Example 6.7

An amplifier has a open-loop voltage gain $A = 1000 \pm 100$. It is required to have an amplifier whose voltage gain varies by no more than $\pm 0.1 \%$.

- Find the reverse transmission factor β of the feedback network used.
- Find gain with feedback.

Solution

Given $A = 1000 \pm 100$

Gain varies between 900 and 1100

Nominal value of gain $A = 1000$

and $\pm \Delta A = \pm 100$

$$\frac{\Delta A}{A} = \frac{100}{1000} = 0.1$$

$$\text{or } \% \frac{\Delta A}{A} = 0.1 \times 100 = 10 \%$$

- It is required to have a gain variation of 0.1%

$$\text{i.e., } \% \frac{\Delta A_f}{A_f} = 0.1 \%$$

$$\text{or } \frac{\Delta A_f}{A_f} = \frac{0.1}{100} = 0.001$$

From Equation (6.79)

$$|1 + \beta A| = \frac{\left| \frac{\Delta A}{A} \right|}{\left| \frac{\Delta A_f}{A_f} \right|} = \frac{0.1}{0.001} = 100$$

$$\beta = \frac{100 - 1}{A} = \frac{99}{1000} = 0.099$$

$$\text{or } \% \beta = 9.9 \%$$

$$(b) \quad A_f = \frac{A}{1 + \beta A} = \frac{1000}{100} = 10$$

$$\text{Now } A_f = 10 \pm 0.1 \% = 10 \pm 0.01$$

Note that A_f varies between 9.99 and 10.01.

Example 6.8

An amplifier consists of three identical stages connected in cascade. The output voltage is sampled and returned to the input in series opposing. It is desired that the relative change $\frac{dA_f}{A_f}$ in the closed-loop voltage gain A_f must not exceed ϕ_f . Show that the minimum value of the open-loop gain A of the amplifier is given by

$$A = [3A_f] \frac{|\phi_1|}{|\phi_f|}$$

where ϕ_1 is defined as $\frac{dA_1}{A_1}$ is the relative change in the voltage gain of each stage of the amplifier.

Solution

$$\left| \frac{dA_f}{A_f} \right| = \frac{\left| \frac{dA}{A} \right|}{|1 + A\beta|} \quad (\text{A})$$

Let A_1 = gain of the each stage without feedback.

$\therefore$ Overall gain without feedback is

$$A = A_1 \times A_1 \times A_1 = A_1^3 \quad (\text{B})$$

Differentiating Equation (B) with respect to A_1

$$\frac{dA}{dA_1} = 3A_1^2$$

$$\text{or} \quad dA = 3A_1^2 \cdot dA_1 \quad (\text{C})$$

From Equation (B)

$$A = A_1^2 \cdot A_1$$

$$\therefore \quad A_1^2 = \frac{A}{A_1}$$

Substituting in Equation (C)

$$dA = \frac{3A}{A_1} \cdot dA_1 \quad (\text{D})$$

$$\text{or} \quad \frac{dA}{A} = 3 \frac{dA_1}{A_1}$$

$$\text{Given} \quad \phi_1 = \frac{dA_1}{A_1}$$

Therefore, Equation (D) becomes

$$\left. \begin{aligned} \frac{dA}{A} &= 3\phi_1 \\ \frac{dA_f}{A_f} &= \phi_f \end{aligned} \right\} \quad \text{(E)}$$

Also given

Substituting Equation (E) in Equation (A)

$$|\phi_f| = \frac{|3\phi_1|}{|1 + A\beta|} \quad \text{(F)}$$

$$\text{But} \quad A_f = \frac{A}{1 + A\beta}$$

$$\therefore 1 + A\beta = \frac{A}{A_f}$$

Substituting this relation in Equation (F)

$$|\phi_f| = 3 \frac{|\phi_1|}{\left| \frac{A}{A_f} \right|} = 3 A_f \frac{|\phi_1|}{|A|}$$

$$\therefore A = 3 A_f \frac{|\phi_1|}{|\phi_f|}$$

Example 6.9

An amplifier without feedback gives a fundamental output of 36 V with 7% second-harmonic distortion when the input is 0.028 V.

- What is the output voltage if 1.2 % of the output is fed back into the input in a negative voltage - series feedback circuit?
- What is the input voltage if the fundamental output is maintained at 36 V but the second harmonic distortion is reduced to 1 %.

Solution

Fundamental output voltage without feedback $V_o = 36 \text{ V}$

% second-harmonic distortion without feedback $\% B_2 = 7 \%$

Input voltage without feedback $V_i = 0.028 \text{ V}$

Voltage gain without feedback

$$\begin{aligned} A_v &= \frac{V_o}{V_i} \\ &= \frac{36}{0.028} = 1285.7 \end{aligned}$$

$$\begin{aligned}
 (a) \quad & \% \beta = 1.2\% \\
 & \therefore \beta = \frac{1.2}{100} = 0.012 \\
 \text{Now } & 1 + A_v \beta = 1 + (1285.7)(0.012) \\
 & = 16.4284 \\
 & A_{vf} = \frac{A_v}{1 + \beta A_v} = \frac{1285.7}{16.4284} \\
 & = 78.26 \\
 & A_{vf} = \frac{V_{of}}{V_{if}}
 \end{aligned}$$

V_{if} is the input voltage with feedback. Since input voltage has not changed,

$$\begin{aligned}
 & V_{if} = V_i = 0.028 \text{ V} \\
 \text{Now } & V_{of} = A_{vf} V_{if} = (78.26)(0.028 \text{ V}) = 2.19 \text{ V} \\
 (b) \quad & \% \text{ second-harmonic distortion with feedback is given as}
 \end{aligned}$$

$$\begin{aligned}
 & \% B_{2f} = 1\% \\
 \text{and } & V_{of} = 36 \text{ V} \\
 & \% B_{2f} = \frac{\% B_2}{1 + \beta A_v} \\
 \therefore & 1 + \beta A_v = \frac{\% B_2}{\% B_{2f}} = \frac{7}{1} = 7
 \end{aligned}$$

Voltage gain with feedback

$$\begin{aligned}
 & A_{vf} = \frac{A_v}{1 + \beta A_v} = \frac{1285.7}{7} = 183.67 \\
 & A_{vf} = \frac{V_{of}}{V_i} \\
 \text{or } & V_i = \frac{V_{of}}{A_{vf}} = \frac{36 \text{ V}}{183.67} = 0.196 \text{ V}
 \end{aligned}$$

Example 6.10

An amplifier with an open-loop gain of 1000 delivers 10 W of output power at 10% second-harmonic distortion when the input signal is 10 mV. If 40 dB negative voltage-series feedback is applied and the output power is to remain at 10 W, determine.

- the required input signal,
- % harmonic distortion

Solution

Open-loop voltage gain means gain without feedback

$$\therefore A_v = 1000$$

Output power without feedback

$$P_o = 10 \text{ W}$$

$$P_o = \frac{V_o^2}{R_L} \quad (\text{A})$$

where R_L is the load resistance.

% second harmonic-distortion without feedback, % $B_2 = 10 \%$

Input voltage, $V_i = 10 \text{ mV}$

$$A_v = \frac{V_o}{V_i}$$

$$\therefore V_o = A_v \cdot V_i = 1000 \times 10 \text{ mV} = 10 \text{ V}$$

Negative feedback in dB,

$$N = -40 \text{ dB}$$

$$N = 20 \log_{10} \left[\frac{1}{1 + \beta A_v} \right]$$

$$= -20 \log_{10} [1 + \beta A_v]$$

$$-40 = -20 \log_{10} [1 + \beta A_v]$$

$$\log_{10} [1 + \beta A_v] = 2$$

$$1 + \beta A_v = \log_{10}^{-1} [2] = 100$$

Now $A_{vf} = \frac{A_v}{1 + \beta A_v} = \frac{1000}{100} = 10$

(a) Output power with feedback is $P_{of} = 10 \text{ W} = P_o$

$$P_{of} = \frac{V_{of}^2}{R_L} \quad (\text{B})$$

Since R_L is fixed, from Equations (A) and (B) we find that

$$V_{of} = V_o = 10 \text{ V}$$

$$A_{vf} = \frac{V_{of}}{V_{if}}$$

$\therefore$ The required input voltage is, $V_{if} = \frac{V_{of}}{A_{vf}} = \frac{10}{10} = 1 \text{ V}$

(b) % second harmonic distortion with feedback is

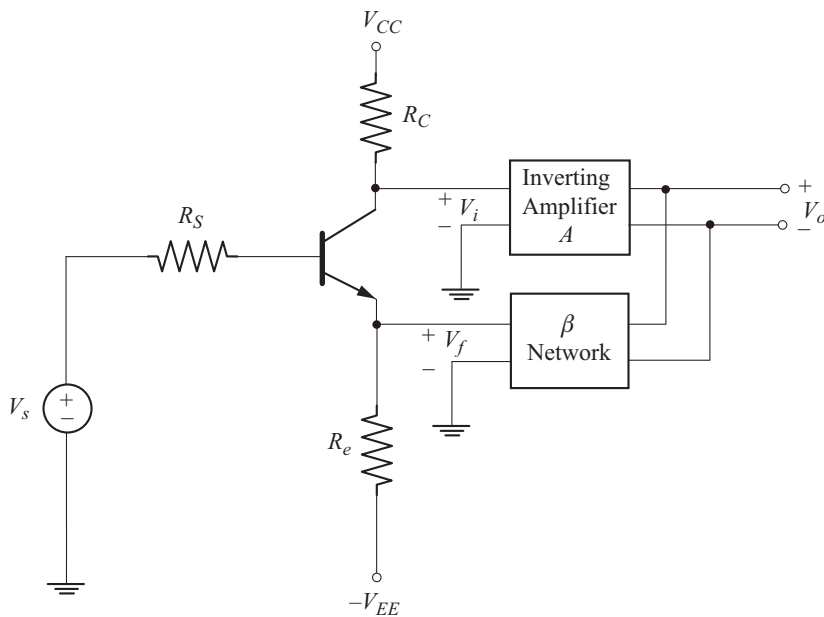
$$\% B_{2f} = \frac{\% B_2}{D} = \frac{10\%}{100}$$

or $\% B_{2f} = 0.1\%$

Example 6.11

(a) For the circuit shown, find the ac voltage V_i as a function of V_s and V_f . Assume that the input resistance of inverting amplifier is infinite. $A = A_v = -1000$, $\beta = \frac{V_f}{V_o} = \frac{1}{100}$, $R_s = R_C = R_e = 1 \text{ k}\Omega$, $h_{ie} = 1 \text{ k}\Omega$, $h_{re} = h_{oe} = 0$ and $h_{fe} = 100$.

(b) Find $A_{vf} = \frac{V_o}{V_s} = \left(A \frac{V_i}{V_s} \right)$



Solution

$$A = A_v = -1000$$

$$\beta = \frac{1}{100} = \frac{V_f}{V_o}$$

$$R_s = R_C = R_e = 1 \text{ k}\Omega$$

$$h_{ie} = 1 \text{ k}\Omega$$

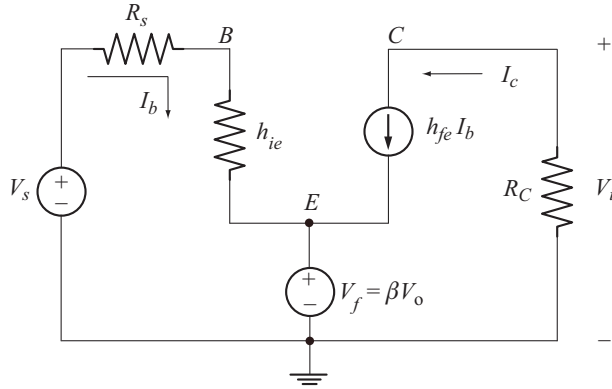
$$h_{re} = h_{oe} = 0$$

$$h_{fe} = 100$$

(a) β network provides an output voltage V_f proportional to its input voltage V_o , which is nothing but the inverting amplifier output voltage.

$$V_f = \beta V_o \quad (\text{A})$$

Let us write the ac equivalent circuit of transistor amplifier. Since $h_{re} = 0$, $h_{re} V_{ce}$ can be treated as short-circuit and $1/h_{oe}$ can be treated as open circuit since $h_{oe} = 0$.



$$V_i = -I_c R_C$$

But

$$I_c = h_{fe} I_b$$

$\therefore$

$$V_i = -h_{fe} I_b R_C \quad (\text{B})$$

Applying KVL to the input circuit

$$V_s - I_b R_s - I_b h_{ie} - V_f = 0$$

$\therefore$

$$I_b = \frac{V_s - V_f}{R_s + h_{ie}} \quad (\text{C})$$

Combining Equations (B) and (C), we get

$$V_i = -h_{fe} R_C \frac{V_s - V_f}{R_s + h_{ie}}$$

$$= -(100) (1 \text{ k}\Omega) \frac{(V_s - V_f)}{(1 \text{ k}\Omega + 1 \text{ k}\Omega)}$$

$\therefore$

$$V_i = -50 (V_s - V_f) \quad (\text{D})$$

(b) For the inverting amplifier

$$V_o = A V_i = A_v V_i$$

Substituting for V_i from Equation (D), we have

$$V_o = A_v [-50 (V_s - V_f)]$$

$$V_o = -50 A_v V_s + 50 A_v V_f$$

Substituting for V_f from Equation (A) we have

$$V_o = -50 A_v V_s + 50 A_v (\beta V_o)$$

$$V_o [1 - 50 \beta A_v] = -50 A_v V_s$$

$$\begin{aligned} \therefore A_{vf} &= \frac{V_o}{V_s} = \frac{-50 A_v}{1 - 50 \beta A_v} \\ &= \frac{-(50)(-1000)}{1 - 50 \left(\frac{1}{100} \right) (-1000)} \\ A_{vf} &= 99.8 \end{aligned}$$

Example 6.12

A voltage amplifier has the following parameter values without feedback.

$$A_v = -1000, \quad R_i = 20 \text{ k}\Omega, \quad R_o = 15 \text{ k}\Omega, \quad \text{Bandwidth} = 200 \text{ kHz}$$

Compute these parameter values if negative-series feedback with $\beta = -0.1$ is given.

Solution

$$1 + \beta A_v = 1 + (-0.1)(-1000) = 101$$

$$A_{vf} = \frac{A_v}{1 + \beta A_v} = \frac{-1000}{101} = -9.9$$

Series mixing input resistance

$$\therefore R_{if} = R_i [1 + \beta A_v] = (20 \text{ k}\Omega)(101) = 2.02 \text{ M}\Omega$$

Voltage sampling reduces output resistance

$$\therefore R_{of} = \frac{R_o}{1 + \beta A_v} = \frac{15 \text{ k}\Omega}{101} = 148.51 \Omega$$

$$BW_f = BW [1 + \beta A_v] = (200 \text{ kHz})(101) = 20.2 \text{ MHz}$$

◆ 6.12 PRACTICAL FEEDBACK CIRCUITS

So far we have studied the effect of negative feedback in various topologies using the block diagram approach. Now let us study the practical feedback circuits.

6.12.1 FET Amplifier with Voltage-series Feedback

Figure 6.30 shows common source FET amplifier with voltage-series feedback.

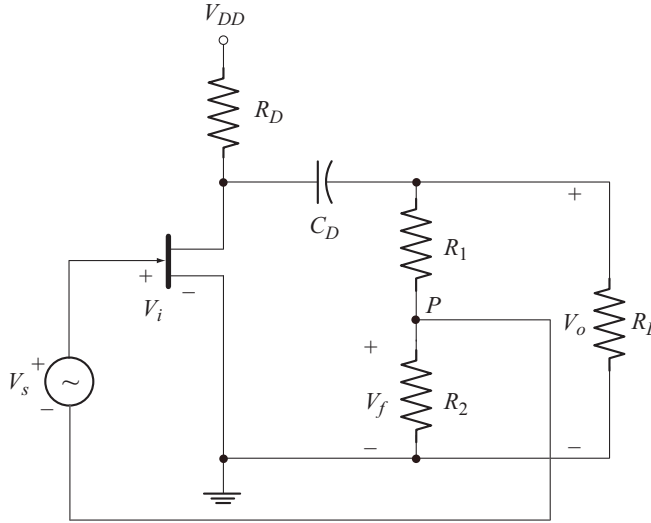


Fig. 6.30 FET amplifier with voltage-series feedback

The output voltage V_o is divided in the voltage divider consisting of R_1 and R_2 . The feedback voltage is the voltage across R_2 . Using voltage division rule,

$$V_f = \frac{R_2}{R_1 + R_2} V_o \quad (6.133)$$

Equation (6.133) is of the form

$$V_f = \beta V_o \quad (6.134)$$

We find that

$$\beta = \frac{V_f}{V_o} = \frac{R_2}{R_1 + R_2} \quad (6.135)$$

Note that the feedback voltage is proportional to output voltage. Hence the sampled signal is the output voltage.

Applying KVL to the input circuit we have

$$\begin{aligned} V_s - V_i - V_f &= 0 \\ \text{or} \quad V_i &= V_s - V_f \end{aligned} \quad (6.136)$$

Note that V_f is mixed in series with V_s . Also V_f subtracts from V_s which implies that feedback is negative.

The sampled signal is the output voltage and the feedback voltage derived from the output voltage is mixed in series with the source voltage. Hence the topology is voltage voltage series feedback. Also the basic amplifier is a voltage amplifier.

Basic Amplifier without Feedback

Let us obtain the ac equivalent circuit of the basic amplifier without feedback from the feedback amplifier of Fig. 6.30.

- To obtain ac equivalent circuit let us short the coupling capacitor C_D and reduce V_{DD} to zero.
- To reduce feedback to zero, we have to open the connection at P and connect it to ground. The ac equivalent circuit of basic amplifier without feedback is shown in Fig. 6.31.

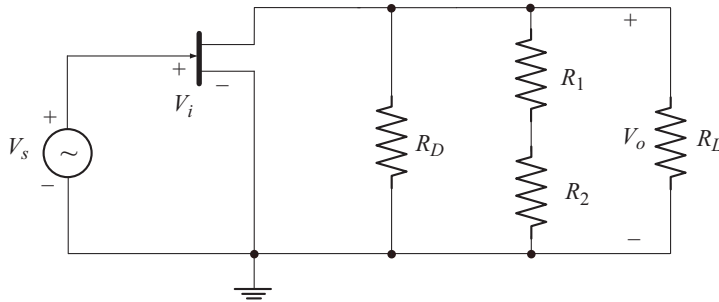


Fig. 6.31 Basic amplifier without feedback

From Fig. 6.31, we find that

$$V_i = V_s \quad (6.137)$$

The effective load on the FET amplifier is

$$R'_L = R_D \parallel [R_1 + R_2] \parallel R_L \quad (6.138)$$

Analysis of FET amplifier is discussed in Chapter 9. The voltage gain of the common-source FET amplifier of Fig. 6.31 is given by

$$A_v = -g_m R'_L \quad (6.139)$$

Voltage gain with feedback is given by

$$A_{vf} = \frac{A_v}{1 + \beta A_v} \quad (6.140)$$

It is important to note that the product βA_v should be positive. Since A_v is negative, it is appropriate to write

$$\beta = \frac{-R_2}{R_1 + R_2} \quad (6.141)$$

Now

$$1 + \beta A_v = 1 + \left[\frac{-R_2}{R_1 + R_2} \right] [-g_m R'_L]$$

$$= 1 + g_m \frac{R_2 R'_L}{R_1 + R_2} \quad (6.142)$$

Using Equations (6.139) and (6.142) in Equation (6.140) we have

$$A_{vf} = \frac{-g_m R'_L}{1 + g_m \frac{R_2 R'_L}{R_1 + R_2}} \quad (6.143)$$

If $\beta A_v \gg 1$, from Equation (6.140) we get

$$A_{vf} \approx \frac{1}{\beta}$$

or

$$A_{vf} = -\frac{R_1 + R_2}{R_2} \quad (6.144)$$

Note that A_{vf} depends only on the external resistors R_1 and R_2 and independent of the FET parameter g_m . If R_1 and R_2 are stable resistors, then A_{vf} is also stabilized.

Example 6.13

The FET amplifier of Fig. 6.30 has the following parameter values.

$$R_1 = 100 \text{ k}\Omega \quad R_2 = 10 \text{ k}\Omega \quad R_L = 15 \text{ k}\Omega$$

$$R_D = 10 \text{ k}\Omega \quad g_m = 5000 \text{ }\mu\text{S}$$

Calculate the voltage gains A_v and A_{vf} .

Solution

$$R'_L = R_D \parallel [R_1 + R_2] \parallel R_L$$

$$= 10 \text{ k}\Omega \parallel [100 \text{ k}\Omega + 10 \text{ k}\Omega] \parallel 15 \text{ k}\Omega$$

$$= 5.68 \text{ k}\Omega$$

$$A_v = -g_m R'_L$$

$$= -(5000 \text{ }\mu\text{S})(5.68 \text{ k}\Omega) = -28.4$$

$$\beta = \frac{-R_2}{R_1 + R_2} = -\frac{10 \text{ k}\Omega}{100 \text{ k}\Omega + 10 \text{ k}\Omega} = -0.09$$

$$1 + \beta A_v = 1 + (-0.09)(-28.4) = 3.556$$

$$A_{vf} = \frac{A_v}{1 + \beta A_v} = -\frac{28.4}{3.556} = -7.9865$$

Example 6.14

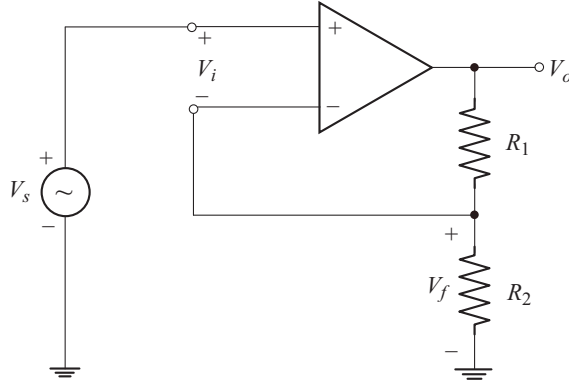
For the op-amp non inverting amplifier shown below, the following data are available:

$$A_v = 2 \times 10^5 \quad R_1 = 10 \text{ k}\Omega$$

$$R_i = 2 \text{ M}\Omega \quad R_2 = 1 \text{ k}\Omega$$

$$R_o = 75 \text{ }\Omega$$

Calculate A_{vf} , R_{if} and R_{of} .



Solution

Using voltage division rule at the output we have

$$V_f = \frac{R_2}{R_1 + R_2} V_o = \beta V_o$$

Note that feedback voltage is derived from output voltage.

$$\begin{aligned} \therefore \beta &= \frac{R_2}{R_1 + R_2} \\ &= \frac{1 \text{ k}\Omega}{10 \text{ k}\Omega + 1 \text{ k}\Omega} = 0.09 \end{aligned}$$

Applying KVL to the input circuit we have

$$\begin{aligned} V_s - V_i - V_f &= 0 \\ \Rightarrow V_i &= V_s - V_f \end{aligned}$$

Note that mixing is series.

Hence the topology is voltage series feedback.

$$1 + \beta A_v = 1 + (0.09)(2 \times 10^5) = 18,001$$

$$A_{vf} = \frac{A_v}{1 + \beta A_v} = \frac{2 \times 10^5}{18,001} = 11.11$$

Voltage sampling reduces the output resistance.

$$\therefore R_{of} = \frac{R_o}{1 + \beta A_v} = \frac{75 \text{ }\Omega}{18,001} = 4.16 \text{ m}\Omega$$

Series mixing increases the input resistance.

$$\therefore R_{if} = R_i [1 + \beta A_v] = (2 \text{ M}\Omega) (18,001) = 36,002 \text{ M}\Omega$$

Note: For op-amp non-inverting amplifier

$$A_{vf} \approx 1 + \frac{R_1}{R_2} = 1 + \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega} = 11$$

This result is consistent with the one which is calculated earlier since $\beta A_v \gg 1$.

6.12.2 Voltage Series Feedback in Emitter-follower

Figure 6.32 shows the circuit of emitter-follower which employs voltage series feedback.

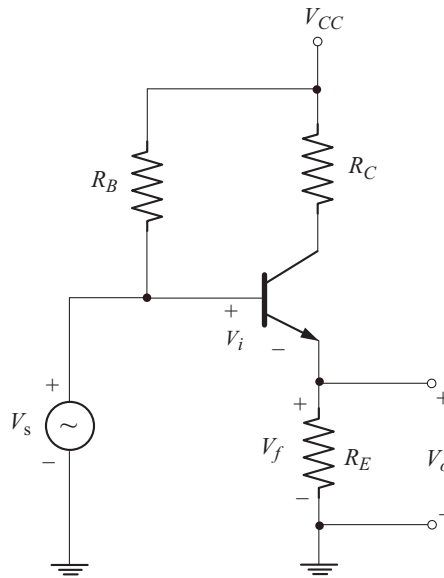


Fig. 6.32 Emitter-follower

From the circuit of Fig. 6.32 we find that

$$V_f = V_o \quad (6.145)$$

$$\text{But } V_f = \beta V_o \quad (6.146)$$

Comparing Equations (6.145) and (6.146) we find that

$$\beta = \frac{V_f}{V_o} = 1 \quad (6.147)$$

Note that the entire output voltage is feedback at the input or the feedback is 100 %.

Applying KVL to the input circuit we have

$$V_s - V_i - V_f = 0$$

$$\text{or} \quad V_i = V_s - V_f \quad (6.148)$$

Note that V_f is mixed in series with V_s . Also, V_f subtracts from V_s which implies that feedback is negative.

The sampled signal is the output voltage and the feedback voltage derived from the output voltage is mixed in series with the source voltage. Hence the topology is voltage series feedback. Also the basic amplifier is a voltage amplifier.

The ac equivalent circuit of emitter-follower is obtained by reducing V_{CC} to zero in the circuit of Fig. 6.32. This is shown in Fig. 6.33.

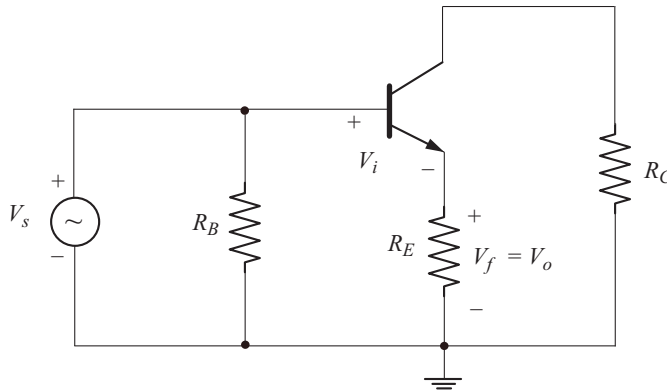


Fig. 6.33 AC equivalent circuit of emitter-follower

Basic Amplifier Circuit without Feedback

Now let us obtain the basic amplifier without feedback. This can be obtained by combining the features of input and output circuits. The procedures for drawing the input and output circuits are given below.

To draw the input circuit, we have to reduce the sampling to zero. Since the output voltage is sampled, it can be reduced to zero by short circuiting R_E in the circuit of Fig. 6.33. The input circuit is shown in Fig. 6.34.

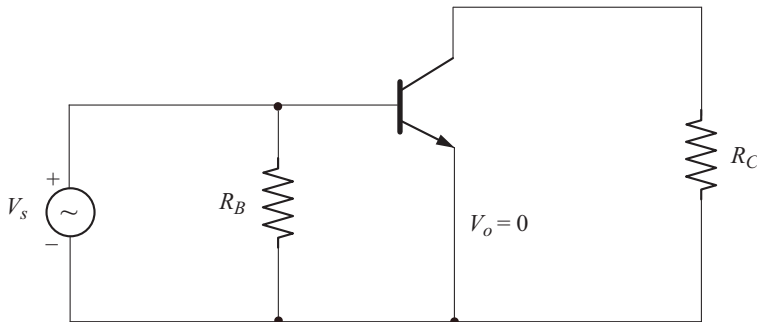


Fig. 6.34 Input circuit obtained by short circuiting R_E

To draw the output circuit, we have to avoid the mixing of V_f and V_s . This can be done by opening the input loop in the circuit of Fig. 6.33. The output circuit is shown in Fig. 6.35.

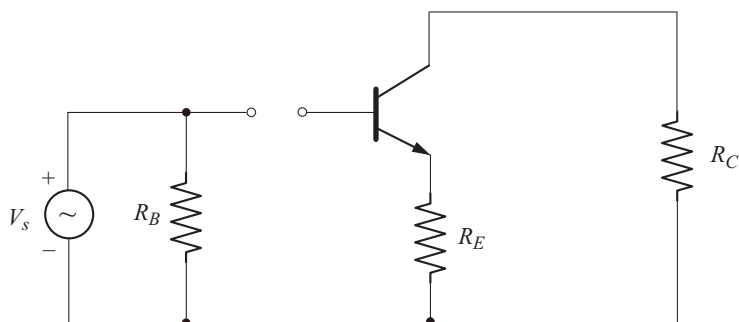


Fig. 6.35 Output circuit obtained by opening the input loop

The basic amplifier circuit without feedback is obtained by combining the features of input and output circuits as shown in Fig. 6.36.

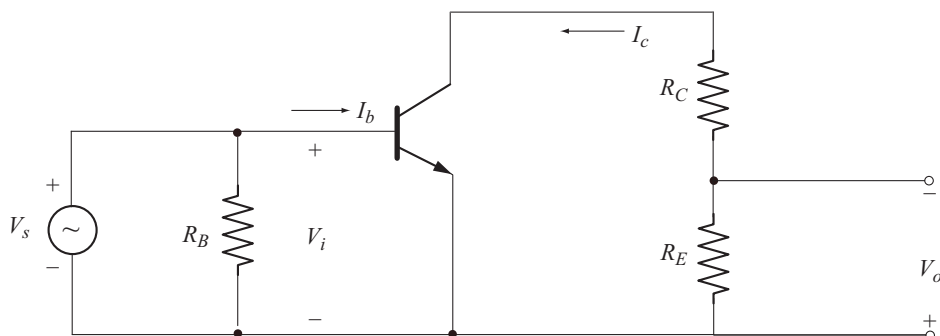


Fig. 6.36 Basic amplifier without feedback

Let us replace the transistor by its low frequency small signal approximate hybrid model as shown in Fig. 6.37.

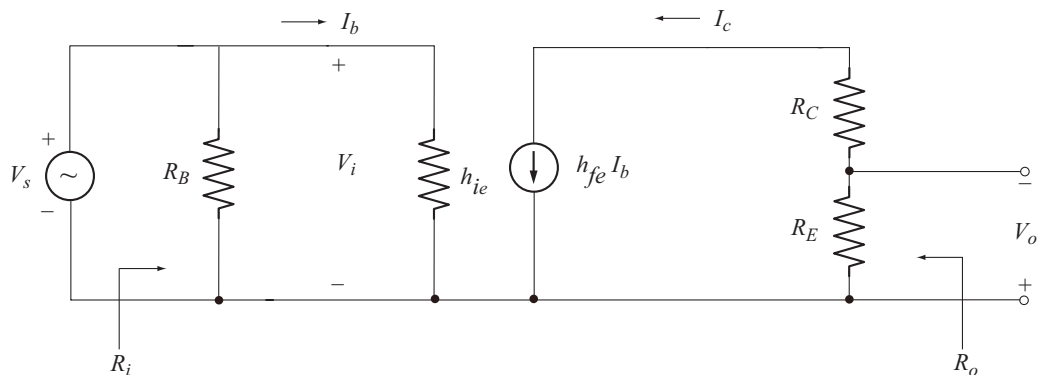


Fig. 6.37 Basic amplifier without feedback using hybrid model

Voltage Gain with Feedback

From the equivalent circuit of Fig. 6.37

$$V_i = h_{ie} I_b \quad (6.149)$$

$$V_o = I_c R_E$$

But $I_c = h_{fe} I_b$

$$\therefore V_o = h_{fe} I_b R_E \quad (6.150)$$

Voltage gain without feedback is

$$\begin{aligned} A_V &= \frac{V_o}{V_i} \\ &= \frac{h_{fe} I_b R_E}{h_{ie} I_b} \\ A_V &= \frac{h_{fe} R_E}{h_{ie}} \end{aligned} \quad (6.151)$$

$$\begin{aligned} 1 + \beta A_V &= 1 + (1) \left[\frac{h_{fe} R_E}{h_{ie}} \right] \\ &= \frac{h_{ie} + h_{fe} R_E}{h_{ie}} \end{aligned} \quad (6.152)$$

Voltage gain with feedback is

$$\begin{aligned} A_{Vf} &= \frac{A_V}{1 + \beta A_V} = \frac{\left[\frac{h_{fe} R_E}{h_{ie}} \right]}{\left[\frac{h_{ie} + h_{fe} R_E}{h_{ie}} \right]} \\ A_{Vf} &= \frac{h_{fe} R_E}{h_{ie} + h_{fe} R_E} \end{aligned} \quad (6.153)$$

Usually

$$\begin{aligned} h_{fe} R_E &\gg h_{ie} \\ \therefore A_{Vf} &\approx \frac{h_{fe} R_E}{h_{fe} R_E} = 1 \end{aligned}$$

which is true for emitter follower.

Input Resistance with Feedback

From the circuit of Fig. 6.37, the input resistance without feedback is

$$R_i = R_B \parallel h_{ie} \quad (6.154)$$

Usually $R_B \gg h_{ie}$

$$\therefore R_i \approx h_{ie} \quad (6.155)$$

Series mixing increases input resistance. Therefore input resistance with feedback is

$$\begin{aligned}
 R_{if} &= R_i [1 + \beta A_v] \\
 &= [h_{ie}] \left[\frac{h_{ie} + h_{fe} R_E}{h_{ie}} \right] \\
 R_{if} &= h_{ie} + h_{fe} R_E
 \end{aligned} \tag{6.156}$$

Output Resistance with Feedback

To find the output resistance without feedback, let us reduce the current source to zero in the output circuit of Fig. 6.37. The resulting circuit is shown in Fig. 6.38.

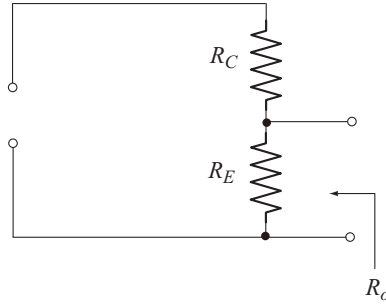


Fig. 6.38 Circuit to find output resistance

From the circuit of Fig. 6.38, output resistance without feedback is

$$R_o = R_E \tag{6.157}$$

Voltage sampling reduces output resistance. Therefore output resistance with feedback is

$$\begin{aligned}
 R_{of} &= \frac{R_o}{1 + \beta A_v} \\
 &= \frac{R_E}{\left[\frac{h_{ie} + h_{fe} R_E}{h_{ie}} \right]} \\
 &= \frac{R_E h_{ie}}{h_{ie} + h_{fe} R_E}
 \end{aligned} \tag{6.158}$$

If $h_{fe} R_E \gg h_{ie}$

then $R_{of} \approx \frac{R_E h_{ie}}{h_{fe} R_E}$

or $R_{of} = \frac{h_{ie}}{h_{fe}}$ (6.159)